



ಸುವರ್ಣ ಭಾರತೀ ಮಹೋತ್ಸವ

ಶ್ರೀಶ್ರೀ ಜಗದ್ಗುರು ಶಂಕರಾಚಾರ್ಯ ಮಹಾಸಂಸ್ಥಾನ
ದಕ್ಷಿಣಾಮ್ನಾಯ ಶ್ರೀಶಾರದಾಪೀಠ, ಶೃಂಗೇರಿ

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WORKSHOP ON THE OCCASION OF SUVARNA BHARATI MAHOTSAVA HANDS ON WORKSHOP ON AI/ML FOR INTERDISCIPLINARY SCIENCES

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Numbers Never Lie... But They Do Confuse!

1. Add 8.563 and 4.8292
2. Find the area of a square whose side is 11 units.
3. Find the value of $x^2 - 5x + 8$, for $x = 2$
4. What should come next in the following letter sequence?

A A B A B C A B C D A B C D E A B C D

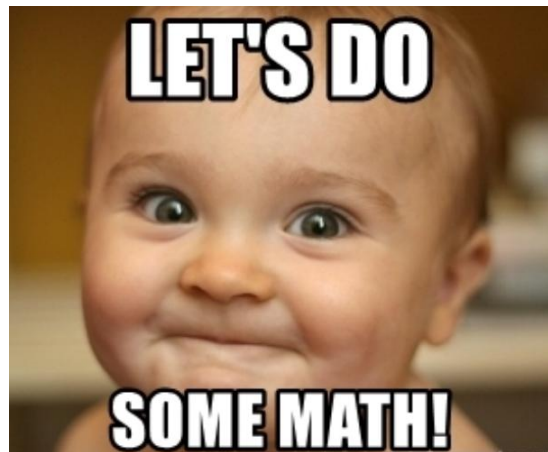
i) A

ii) E

iii) C

iv) B

5. 1, 3, 5, 7, ?
6. 24, 26, 28, ?, 32, 34



$$^{\circ}\text{F} = [^{\circ}\text{C} \times (9/5)] + 32$$

1. Degree Celsius to Fahrenheit

Temperature (°C)	Calculation	Temperature (°F)
0°C	$(0 \times 9/5) + 32$	32°F
10°C	$(10 \times 9/5) + 32$	50°F
20°C	$(20 \times 9/5) + 32$	68°F
30°C	$(30 \times 9/5) + 32$	86°F

2. Body Mass Index

Weight (kg)	Height (m)	BMI Value	Category
50	1.6	19.53	Normal weight
68	1.75	22.2	Normal weight
85	1.7	29.41	Overweight
110	1.75	35.92	Obese
45	1.5	20	?
95	1.8	?	?



Does our Natural Intelligence stop working without an equation?



If so, how do our mothers cook without a recipe?

We **learn from experience**, adjust ingredients, and use **intuition**—that's Natural Intelligence!

But what if something **didn't learn from experience** and **only followed instructions**?

That's where **Artificial Intelligence** comes in!

When we calculate **BMI**, we don't guess—it follows a **fixed formula**

$$\text{Body Mass Index} = \frac{\text{Weight (kg)}}{\text{Height(m}^2\text{)}}$$

What is Artificial Intelligence?



- Just like our mothers cook without a fixed recipe, AI doesn't need an equation, it learns patterns from data!
- AI doesn't need to be given this formula—it analyzes thousands of weight and height records and finds the pattern on its own.
- AI sees that as weight increases, BMI usually goes up.
- AI notices that taller people with the same weight tend to have a lower BMI.
- It doesn't “know” the equation, but it figures out the relationship just like how we learn to cook by observing!
- That's AI—it finds patterns, makes predictions, and helps us make decisions without needing strict rules!

Defining Artificial Intelligence?

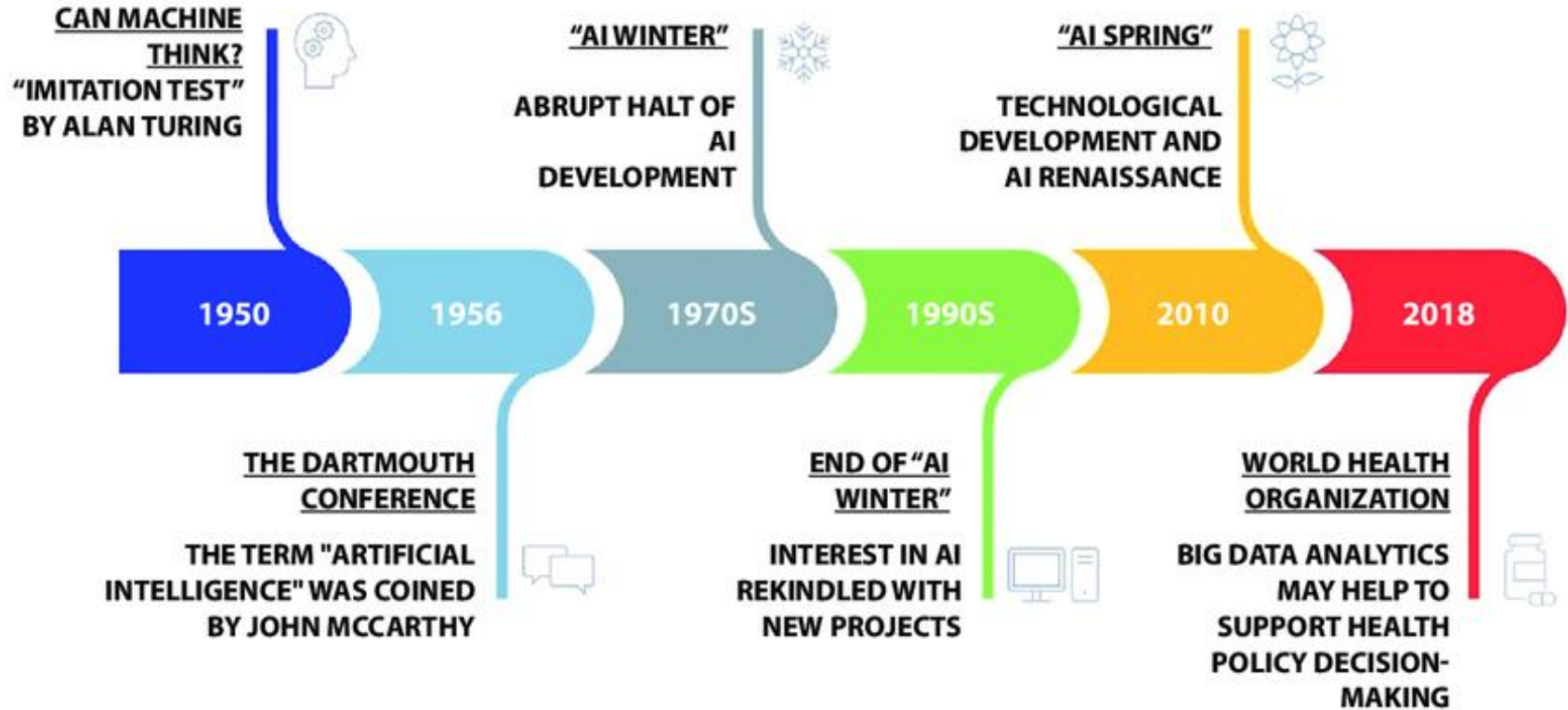
It is the ability of machines to learn, think, and make decisions like humans—but using data and patterns instead of intuition. Unlike us, AI doesn't need to “understand” concepts—it finds relationships from large amounts of information.

- **How AI Works?**

- ◆ Learns from Data  (instead of experience like humans)
- ◆ Finds Patterns  (instead of memorizing rules)
- ◆ Makes Predictions & Decisions  (based on past examples)

Is AI New?

TIMELINE DIAGRAM OF ARTIFICIAL INTELLIGENCE HISTORY



Data And Its Importance For AI



- Just like a car can't run without fuel, AI can't learn, think, or make decisions without data.
- AI Learns from Data, Not Experience.
- It learns by analyzing huge amounts of data to recognize patterns and trends.
- If we give AI thousands of weight and height records, it learns the relationship between them and predicts BMI.
- AI without data is like a student without books, a chef without ingredients.

Learning Theory in AI?



Learning Theory in AI is like teaching a child how to recognize patterns and improve over time. Instead of memorizing everything, AI learns from experience (data) and gets better with practice.

<https://youtu.be/8ppLbs13JMY?si=9K9TZv0IO6cJtGok>

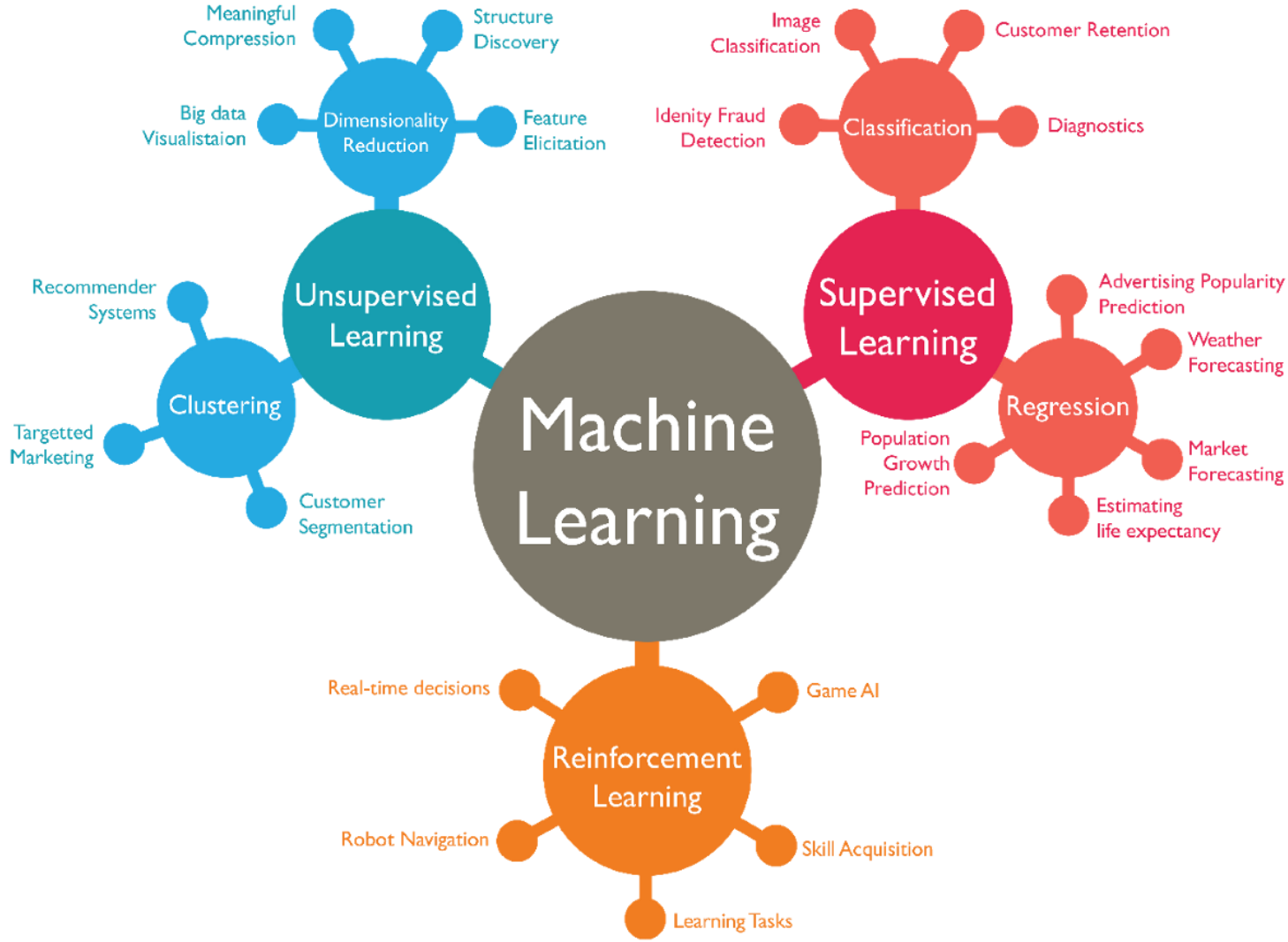
Different Types of Tasks AI Can Perform

- Classification – Sorting Things into Categories
- Prediction – Forecasting Future Outcomes
- Pattern Recognition – Finding Hidden Relationships
- Image Analysis – Understanding Medical Images
- Drug Discovery – Finding New Medicines Faster

Role of AI In Different Domains



1. **Healthcare** – AI assists in medical diagnosis, drug discovery, robotic surgeries, and personalized treatment plans.
2. **Agriculture** – AI-driven predictive analytics, smart irrigation, and crop monitoring enhance productivity and sustainability.
3. **Finance** – AI automates fraud detection, algorithmic trading, risk assessment, and customer service (chatbots).
4. **Education** – AI-powered personalized learning, automated grading, and intelligent tutoring systems improve teaching efficiency.
5. **Retail** – AI enhances customer experience through recommendation systems, demand forecasting, and automated checkouts.



Linear Regression Exercise -1

Lets make Machines Learn



Linear Regression is a supervised learning technique used for predicting continuous numerical values based on input features. It establishes relationships between independent and dependent variables. It helps us predict one value based on another by fitting a straight line through the data points.

Linear Regression – Models the relationship between input (X) and output (Y) using a straight-line equation:

$$Y=mX+C$$

Hands –on Exercise

Predicting Blood Pressure from Age

Let's say we want to understand how age affects blood pressure. We collect data from different people and observe that as age increases, blood pressure also tends to increase.

Age (Years)	Blood Pressure (mmHg)
20	110
25	115
30	118
35	122
40	126
45	130
50	135

- **Plot the Data Points**

We place age on the X-axis and blood pressure on the Y-axis and see a general trend: older people tend to have higher blood pressure.

- **Find the Best-Fitting Line**

AI draws a straight line that best represents this trend. This line is called the regression line, and its equation is:

$$\text{Blood Pressure} = m(\text{Age}) + b$$

m = slope (how much blood pressure changes per year of age)

b = intercept (blood pressure when age is 0)

- **Make Predictions**

Now, if we want to predict the blood pressure of a 55-year-old, we simply use the equation!

For example, if AI finds that:

$$\text{Blood Pressure} = 0.5 \times \text{Age} + 105$$

for Age = 55, we get:

$$\text{Blood Pressure} = 0.5(55) + 105 = 132.5 \text{ mmHg}$$

So, we predict that a 55-year-old will likely have a blood pressure of 132.5 mmHg.

Machine Learning Clustering Example In Microbiology

Let's consider a scenario where we group bacteria based on their growth rate and temperature preference.

How K-Means Works Here?

Step 1: Select the number of clusters (k)

Let's choose $k = 2$

(we expect two groups: slow-growing vs. fast-growing bacteria).

Step 2: Assign Initial Cluster Centers

AI picks two random bacteria as starting cluster centers.

Step 3: Assign Each Bacteria to the Nearest Cluster

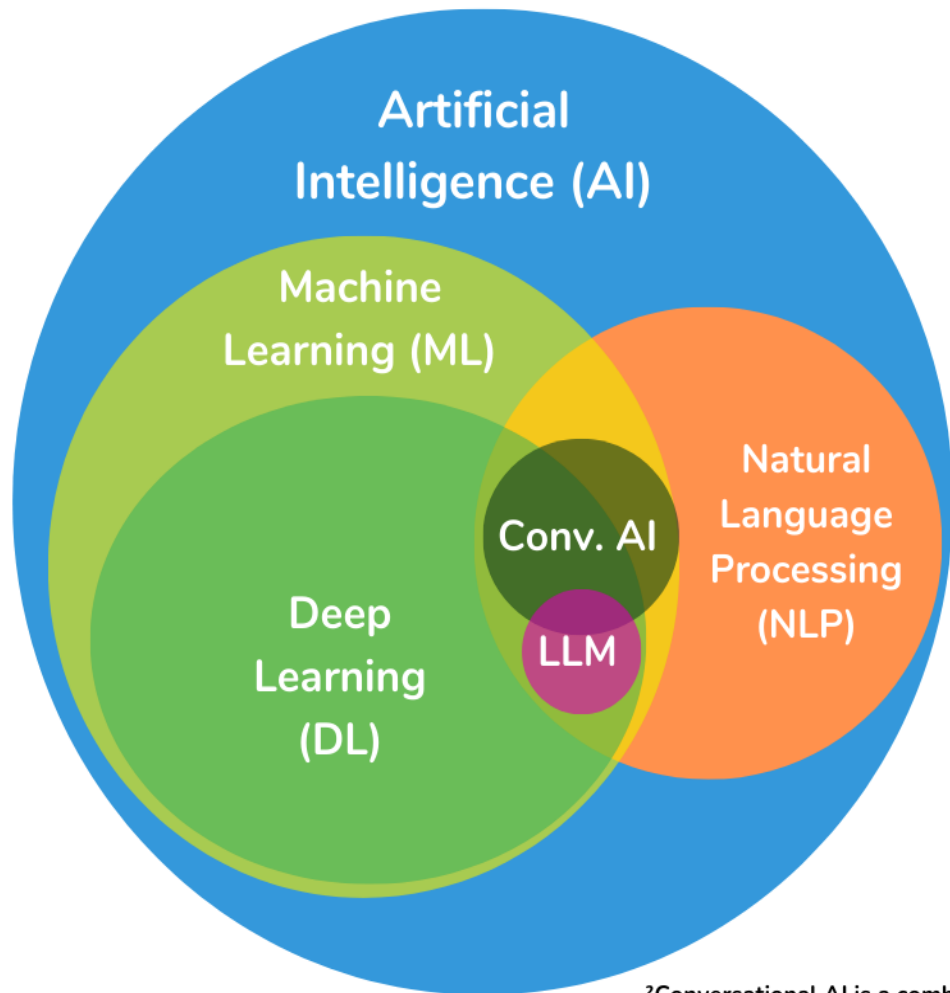
Bacteria with similar growth rates and temperature preferences will be grouped together.

Step 4: Update the Centroids

AI recalculates the average of each cluster and reassigns bacteria until stable clusters form.

Bacteria	Growth Rate (per hour)	Optimal Temperature (°C)
A	0.5	25
B	0.7	30
C	1.2	37
D	1.3	40
E	1.5	45
F	0.4	22

Post Lunch Session



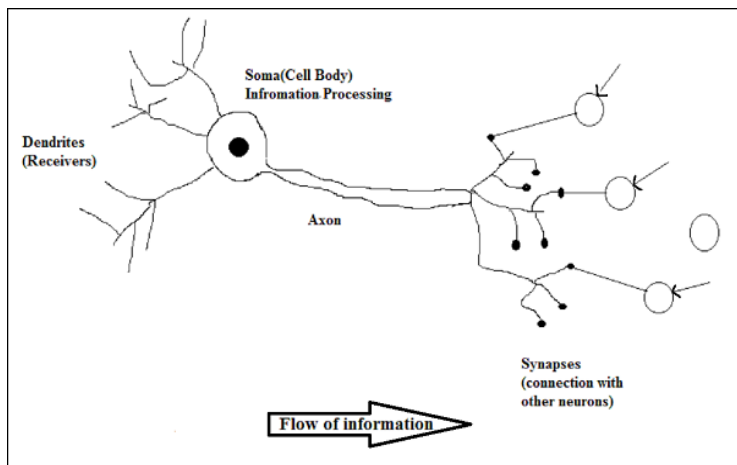
- Artificial Intelligence (AI)
- Machine Learning (ML)
- Deep Learning (DL)
- Natural Language Processing (NLP)
- Large Language Model (LLM)¹
- Conversational AI (Conv. AI)²

¹LLM is an intersection of DL and NLP

²Conversational AI is a combination of ML and NLP. It may include DL and LLM, but that isn't always the case.

Deep Learning

Biological Neural Network BNN	Artificial Neural Network ANN
Soma	Node
Dendrites	Input
Synapse	Weights or Interconnections
Axon	Output



Criteria	BNN	ANN
	Massively parallel, slow but superior than ANN	Massively parallel, fast but inferior than BNN
Size	10^{11} neurons and 10^{15} interconnections	10^2 to 10^4 nodes mainly depends on the type of application and network design.
Learning	They can tolerate ambiguity	Very precise, structured and formatted data is required to tolerate ambiguity
Fault tolerance	Performance degrades with even partial damage	It is capable of robust performance, hence has the potential to be fault tolerant
Storage capacity	Stores the information in the synapse	Stores the information in contiguous memory locations

Computer Vision is like how **our eyes and brain work together** to understand the world.
Its like a child recognizing fruits.

- **Capturing Images** – A camera or microscope takes a picture.
- **Processing the Image** – The image is cleaned and enhanced (e.g., improving brightness, removing noise).
- **Finding Patterns** – The computer looks for shapes, textures, and colors to recognize objects.
- **Making Predictions** – AI models classify what is in the image (e.g., detecting bacteria or counting cells).

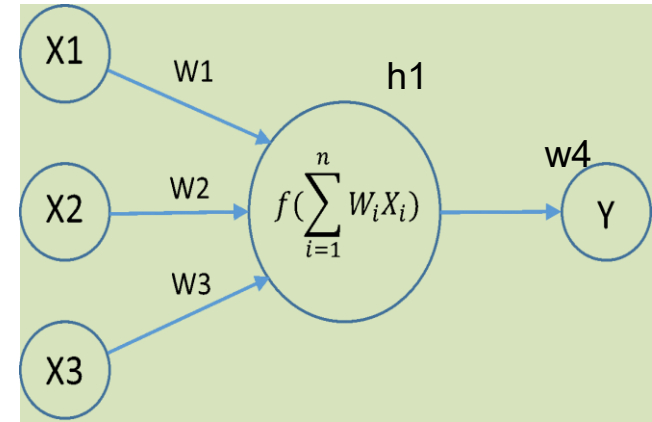
Applications of Computer Vision in Microbiology

1. **Cell and Bacteria Identification** – Detecting and classifying microorganisms in microscopic images.
2. **Disease Diagnosis** – Analyzing medical images (e.g., identifying infected cells).
3. **Colony Counting** – Automating bacterial colony counting in petri dishes.
4. **Quality Control in Labs** – Checking for contamination or abnormalities in samples.
5. **Handwriting Recognition** – Converting lab notes into digital text using Optical Character Recognition (OCR).

Hands –on Exercise

Basics Of Artificial Neural Network

- Consists of three or more layers and multiple neurons in each layer
- Generalized approximation model
- *Neuron*: Processing unit that takes input and produces some output
- *Layer*: One input layer, one output layer and one or more hidden layers



$x = (x_1, \dots, x_n)$ vector of inputs
 $w = (w_1, \dots, w_n)$ vector of weights
 $b = \text{bias}$, $y = \text{output}$

Basics Of Artificial Neural Network

- A neural network is usually described as having different layers. The first layer is the **input layer**, it picks up the input signals and passes them to the next layer.
- The next layer does all kinds of calculations and feature extractions—it's called the **hidden layer**.
- Often, there will be more than one hidden layer.
- There's an **output layer**, which delivers the final result.
- Bias: **Bias** allows you to shift the activation function by adding a constant (i.e. the given **bias**) to the input. **Bias in Neural Networks** can be thought of as analogous to the role of a constant in a linear function.

Backpropagation is a **supervised learning algorithm**, for training Multi-layer Perceptrons (Artificial Neural Networks).

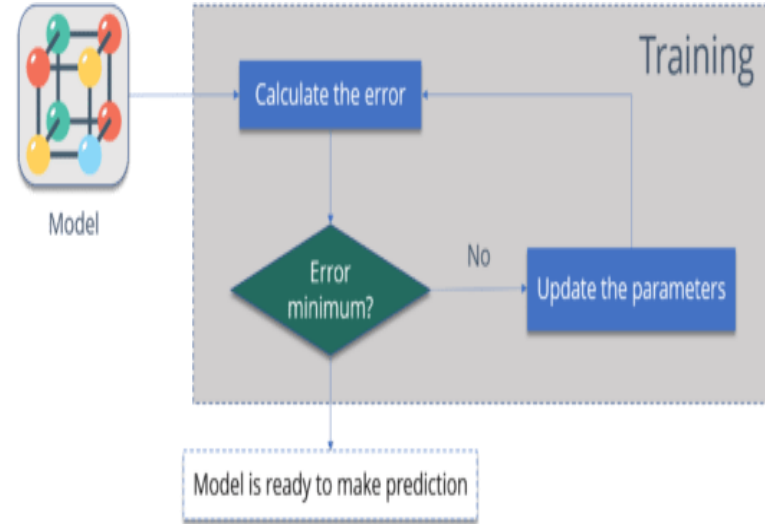
Why We Need Backpropagation?

While designing a Neural Network, in the beginning, we **initialize weights** with **some random values**.

We have selected some weight values in the beginning, but our model output is way different than our actual output i.e. the error value is huge.

Now, how will you reduce the error?

we need to train the model to change the weights, such that error becomes minimum.



What is Backpropagation?

The Backpropagation algorithm looks for the minimum value of the error function in weight space using a technique called the delta rule or gradient descent. The weights that minimize the error function is then considered to be a solution to the learning problem.

Learning Rate=**learning rate** is a configurable hyperparameter used in the training of neural networks that has a small positive value, often in the range between 0.0 and 1.0. The **learning rate** controls how quickly the model is adapted to the problem.

How AI can be Leveraged In Biology

1. Microbiology

- **Bacterial Identification:** AI algorithms can classify bacteria from microscope images or DNA sequences, making the identification process faster and more accurate.
- **Antibiotic Resistance Prediction:** AI models can predict how bacteria will evolve and become resistant to antibiotics, helping in the development of new treatments.
- **Disease Diagnosis:** AI can analyze medical images (e.g., X-rays, MRIs) to detect signs of infection, tumors, or abnormalities in tissues with higher accuracy than human experts.
- **Microbial Genome Analysis:** AI can speed up the process of sequencing and analyzing microbial genomes, aiding in the study of new microorganisms and their properties.

How is Machine Learning Done?

Machine Learning involves several key steps:

1. **Data Collection** – Gathering relevant data, either structured (databases, spreadsheets) or unstructured (text, images).
2. **Data Preprocessing** – Cleaning, normalizing, and transforming data to remove inconsistencies and improve model efficiency.
3. **Model Selection** – Choosing an appropriate algorithm (e.g., regression, classification, clustering) based on the problem type.
4. **Training the Model** – Feeding data into the model to learn relationships and adjust parameters.
5. **Evaluation & Optimization** – Measuring performance using metrics (e.g., accuracy, loss, R^2 score) and fine-tuning for better results.
6. **Deployment & Monitoring** – Using the trained model in real-world applications and continuously improving it with new data.

*Thank
you*

